University of Pretoria
Department of Mathematics and Applied Mathematics

WTW 158 Calculus

ASSIGNMENT: Supplementary Exam July 2021

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Section A: 38 marks (90 minutes)

INSTRUCTIONS:

1. The exam paper is an assignment that must be uploaded.

2. Draw a line with a ruler at the end of each question.

3. Write your surname, initials and student number clearly in the beginning of your assignment.

4. Only handwritten assignments submitted in one PDF file will be accepted.

5. Before submitting, make sure that you are satisfied with your answers and that you submit the correct document.

6. No late assignments will be accepted.

7. No assignments sent via e-mail will be accepted.

8. Calculators **may not** be used.

Section A: 90 minutes

In Question 1, you must write down the FINAL ANSWER ONLY. Only this will be marked. Please do not show us your calculations and rough work.

Question 1 [16 marks]

(1.1) Solve the inequality: $|5-x| \leq 2$. Write down the solution set in interval notation. (1)

(1.2) Find the distance between the point $A\,(3, 1, -4)$ and the y-axis. Write down your answer only. (1)

(1.3) Determine the Riemann-sum approximation of

$$\int_{-1}^{3} x^2 dx$$

using a right-hand sum with 4 subintervals. Write down your final answer only. (1)

(1.4) Write down the equation(s) of the horizontal asymptote(s) of the function (1)

$$f\,(x) = \frac{4x-4}{x^2-1}.$$

(1.5) Suppose that $F(x) = f(g(x))$, $f(2) = 4, f'(3) = 5, g(2) = 3$ and $g'(2) = 4$. Write down the value of $F'(2)$ only. (2)

(1.6) Write down an anti-derivative of (2)

$$\frac{2}{x^2+4}.$$

(1.7) Use interval notation and write down the domain of (2)

$$2\sin^{-1}(x-4).$$

(1.8) Consider the function: (2)

$$f\,(x) = \begin{cases} \frac{\sin 2x}{x} & \text{if } x \neq 0 \\ \ln a & \text{if } x = 0 \end{cases}$$

Write down the value of a for which f will be continuous at $x = 0$.

(1.9) Consider the function (2)

$$f(t) = \ln t - 2t^2.$$

Write down the critical number(s) of f.

(1.10) Write down the equation of the tangent line to the curve (2)

$$y = \frac{1}{1-x}$$

at the point $(2, -1)$.

From here onwards, you must show all steps and calculations made to get to the final answer. Motivate each step and do not skip any steps.

Question 2 [4 marks]

Consider $f(x) = x - 2\sin x$ for $x \in (-\pi, \pi)$.

(2.1) Find the interval(s) where f is increasing. (2)

(2.2) Find the interval(s) where f is concave up. (2)

Question 3 [2 marks]

Find the inverse of the function

$$f(x) = 2\ln(x + a), a > 0.$$

Question 4 [5 marks]

(4.1) Determine the gradient of the tangent line to the curve

$$xy + y^3 = \sec x$$

in the point (x, y). (2)

(4.2) Determine the gradient of the tangent line to the curve

$$y = (x^2 + x)^x$$

at $x = 1$. (3)

Question 5 [4 marks]

Consider the points $A(1, -1, 1)$ and $B(3, 2, -1)$ in the three dimensional space. Find two unit vectors orthogonal to both $\overline{OA}$ and $\overline{OB}$. Verify that these vectors are indeed orthogonal to both $\overline{OA}$ and $\overline{OB}$.

Question 6 [5 marks]

(6.1) Determine (2)

$$\lim_{x \to -\infty} x^2 e^x$$

if it exists.

(6.2) Consider the function (3)

$$f(x) = x^x,$$

which is defined for $x > 0$. Determine

$$\lim_{x \to 0^+} f(x).$$

Question 7 [2 marks]

Find the equation of the line through the origin that is tangent to the function $f(x) = e^x$.